注意：

允許學生個人、非營利性的圖書館或公立學校合理使用本基金會網站所提供之書籍、各項試題及其解答。**可直接下載而不須申請**。

重版、系統地複製或大量重製這些資料的任何部分，必須獲得財團法人臺北市九章數學教育基金會的授權許可。

申請此項授權請電郵 chiuchang.tw@gmail.com

Notice:

Individual students, nonprofit libraries, or schools are permitted to make fair use of the books, papers and its solutions. Republication, systematic copying, or multiple reproduction of any part of this material is permitted only under license from the Chiuchang Mathematics Foundation.

Requests for such permission should be made by e-mailing Mr. Wen-Hsien SUN chiuchang.tw@gmail.com

International Mathematics Assessments for Schools

2025 / 2026 MIDDLE PRIMARY DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

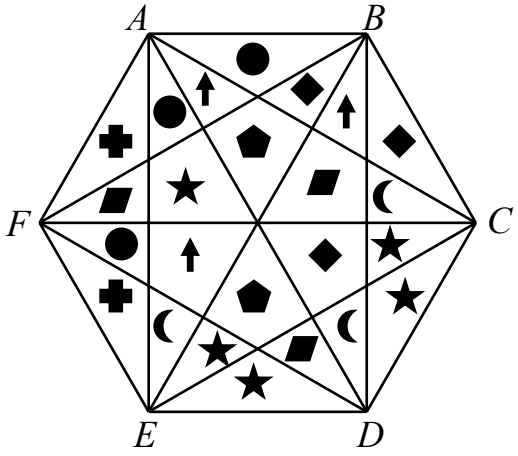
INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2025 / 2026 MIDDLE PRIMARY DIVISION FIRST ROUND PAPER

Questions 1-10, 3 marks each

1. As shown in the diagram below, each region of the hexagon contains exactly one symbol.

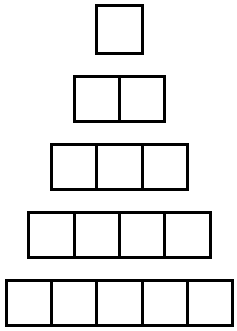


Among the options, which triangle contains a set of symbols where no symbol appears more than once?

(A) *ADF* (B) *BCE* (C) *CEF* (D) *ACD* (E) *BEF*

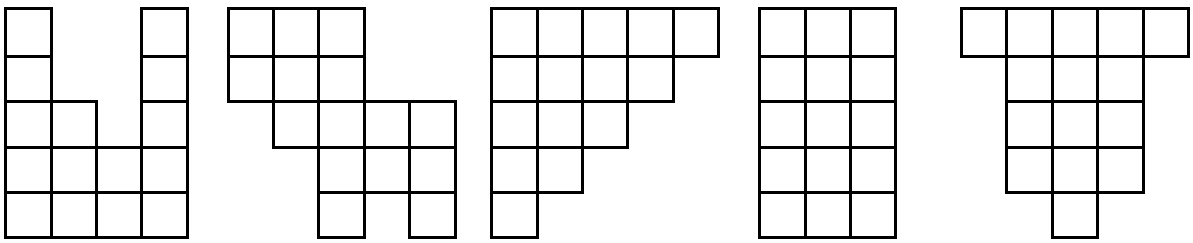
---

2. As shown in the diagram below, Eva has five tiles, each of a different size.



She wants to use all five tiles at the same time, without any overlaps, to form a shape. Among the options, which shape can Eva **NOT** form?

(A) (B) (C) (D) (E)



---

3. Students from three levels, Middle Primary, Upper Primary and Junior, participate in a math competition. If 65% of the students are not in Upper Primary and 60% are not in Middle Primary, then what percentage of students are Junior participants?

（A）20%　（B）25%　（C）30%　（D）35%　（E）40%

---

4. In Braille code, a positive integer is represented by the number sign followed by the Braille symbols for each of its digit, as shown in the table below. For example, the number 23 is represented as ⠼⠃⠉, using a total of 8 dots.

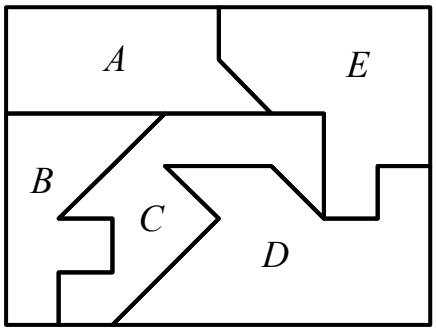
Number sign	1	2	3	4	5	6	7	8	9	0
⠼	⠁	⠃	⠉	⠙	⠑	⠋	⠛	⠓	⠊	⠚

Including the number sign ⠼, how many dots are needed to represent 2025?

（A）9　（B）10　（C）11　（D）12　（E）13

---

5. A rectangle is divided into five regions, as shown in the diagram below.



If the given diagram is precisely drawn to scale, what is the ordering of the areas of the regions, from largest to smallest?

（A）$D > E > A > B > C$　（B）$D > E > C > A > B$　（C）$D > E > B > C > A$

（D）$E > D > C > A > B$　（E）$E > D > A > C > B$

---

6. The IMAS baseball team won 45 of its first 50 games. What is the least number of games they must win out of their remaining 40 games to finish the season with a winning percentage of at least 75%?

（A）18　（B）20　（C）22　（D）23　（E）24

---

7. There are three friends: Anna, Bob and Chris.

Anna says: “Bob is lying!”.
Bob says: “Chris is lying!”.
Chris says: “Anna and Bob are both lying!”.

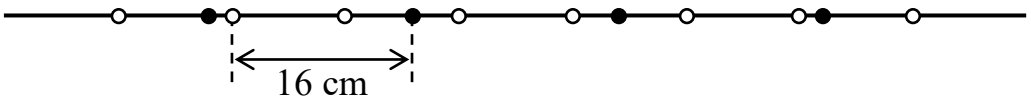
Who among the three is telling the truth?

（A）Anna　（B）Bob　（C）Chris

（D）all of them　（E）neither of them

---

8. A straight stick lies horizontally on the ground. It has four black dots and eight white dots such that the black dots divide the stick into five equal parts and the white dots divide the stick into nine equal parts. If the distance between the second black dot and the second white dot from left to right is 16 cm, as shown in the diagram below, what is the length, in cm, of the stick?



(A) 45 (B) 80 (C) 90 (D) 120 (E) 160

---

9. A taxi driver charges passengers a fixed rate of \$2.50 for any trip that is one kilometer or less. For trips over one kilometer, an additional \$0.15 is charged for every 100 meters beyond the first kilometer. A passenger was charged \$5.95 for a trip. How far did the passenger travel?

(A) 1.23 km (B) 2.3 km (C) 3.3 km (D) 23 km (E) 24 km

---

10. Let $\overline{410a676}$ be a 7-digit number which is a multiple of 2026. What is the value of a?

(A) 0 (B) 2 (C) 4 (D) 6 (E) 8

Questions 11-20, 4 marks each

11. Three friends, Alex, Ben and Charlie, are participating in a 1600–meter inline skating race. The race results are as follows:

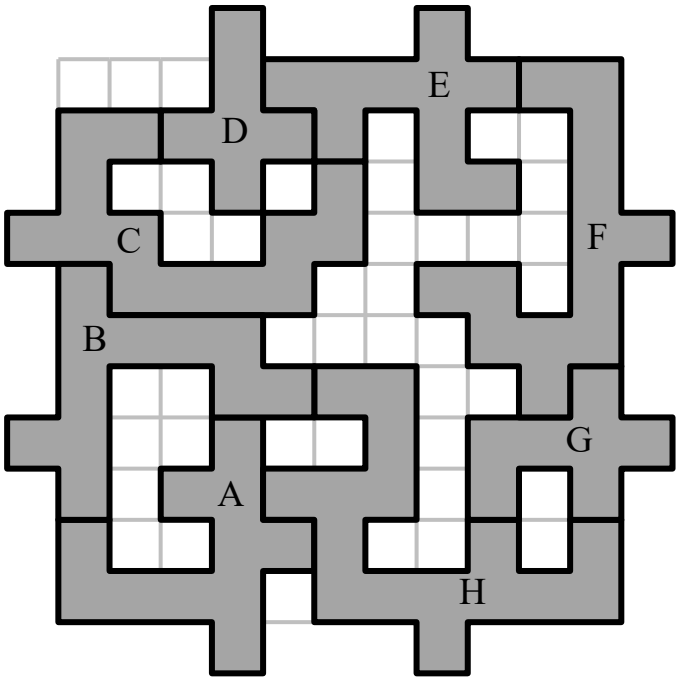
- Alex finishes the race first.
- Ben finishes 18 seconds after Alex.
- When Ben finishes, Charlie still has 120 meters left to skate.
- Charlie takes 15 more seconds to reach the finish line after Ben.

Assuming all three skate at constant speeds, how long, in seconds, did Alex take to finish the race?

(A) 167 (B) 177 (C) 182 (D) 185 (E) 233

---

12. As shown in the diagram below, David has eight shaded pieces (labelled as A, B, …, G, H) that are placed on the 11×11 game board. Each piece is to be pushed, in a certain order, into the game board such that every piece must align along the grid lines and no part of any piece remains outside the board or overlapping. After David pushes all eight shaded pieces in this order, which piece will be pushed last?



(A) A (B) B (C) D (D) G (E) H

---

13. Start with the number 2025 and perform the following operation: every time a 0 appears in the number, replace it with 2025. For example, after the first operation, the number becomes 2202525. If this operation is repeated 2025 times, what is the positive difference between the number of 2s and the number of 5s in the final number?

(A) 0 (B) 1 (C) 1013 (D) 2025 (E) 2026

---

14. The sum of 25 different positive odd integers is 2025. What is the largest possible value among these odd integers?

(A) 1441 (B) 1443 (C) 1445 (D) 1447 (E) 1449

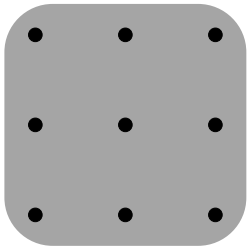
---

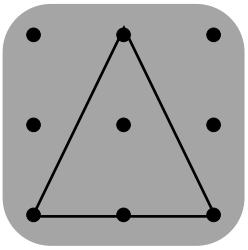
15. There are 24 four-digit positive integers that use each of the digits 2, 4, 5 and 7 exactly once. Only one of these four-digit numbers is a multiple of a different number. Which of the following numbers is it?

(A) 5724 (B) 7245 (C) 7254 (D) 7425 (E) 7524

---

16. As shown in the diagram on the left, a 3×3 grid is created on a wall by using nine nails, with the distance between any two adjacent nails, whether horizontally or vertically, is equal to 10 cm. A rubber band is then used to connect three non-collinear points on the grid to form a triangle. The diagram on the right shows one example of a triangle formed in this manner.





How many different possible value(s) can the area of the resulting triangle take on?

（A）1　（B）2　（C）3　（D）4　（E）5

---

17. A class of students took a test. After marking all the test papers, the teacher decided to award 2 extra points to each boy and 3 extra points to each girl. As a result, the class average increased by 2.2 points. What percentage of the students in the class are girls?

（A）20%　（B）25%　（C）30%　（D）40%　（E）80%

---

18. Alice and Ben are participating in the “Healthy Walk Challenge”, where each participant must walk starting from Sunshine Elementary School to Green Park. When they reach Rest Point 1, Alice asks Ben: “How far have we walked so far?”

Ben replies: “We have walked one-third of the distance from here to Rest Point 2.”

After resting, they continue walking. After walking another 18 kilometers, Alice asks again: “How much farther do we have to walk to reach Green Park? I’m getting tired!”

Ben replies: “Only one-third of the distance from here back to Rest Point 2 remains, and then we will be at Green Park.”

How far apart, in kilometers, are Sunshine Elementary School and Green Park?

（A）20　（B）21　（C）22　（D）23　（E）24

---

19. Consider all four-digit numbers whose product of digits is equal to 0. If these numbers are listed in increasing order, what position does the number 2025 occupy?

（A）295　（B）296　（C）297　（D）298　（E）729

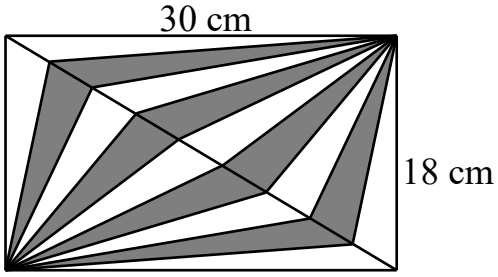
---

20. A frog starts at 0 on a number line. Each jump is either +26 or +20. Is it possible for the frog to land on 2026? If it is not possible, choose option (A). If it is possible, what is the minimum number of jumps required to achieve this?

（A）It is impossible.　（B）71　（C）80
（D）92　（E）101

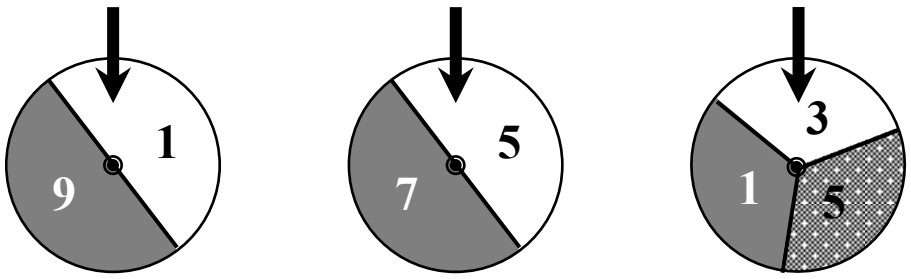
Questions 21-25, 6 marks each

21. As shown in the diagram below, one diagonal of an 18 cm by 30 cm rectangle has been divided into 9 equal segments. What is the total area, in cm^2, of all the shaded regions?



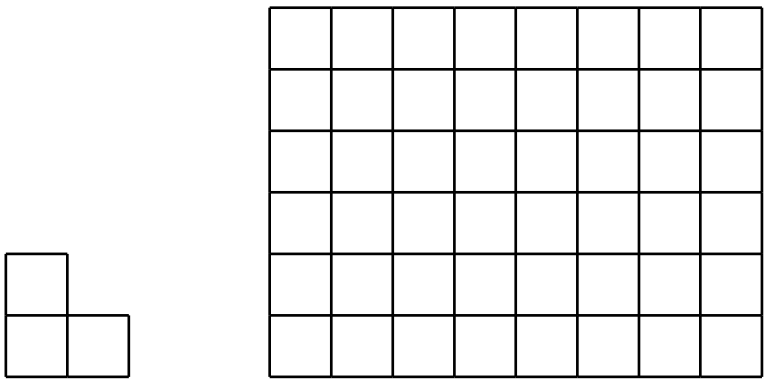
22. Three spinners, shown below, are all spun at the same time, each landing on one of its numbered sectors. In how many ways can the sum of the three numbers be a prime number?

Note: If the sums are the same, but the combinations of the three numbers are different, then each combination is counted as a different case.

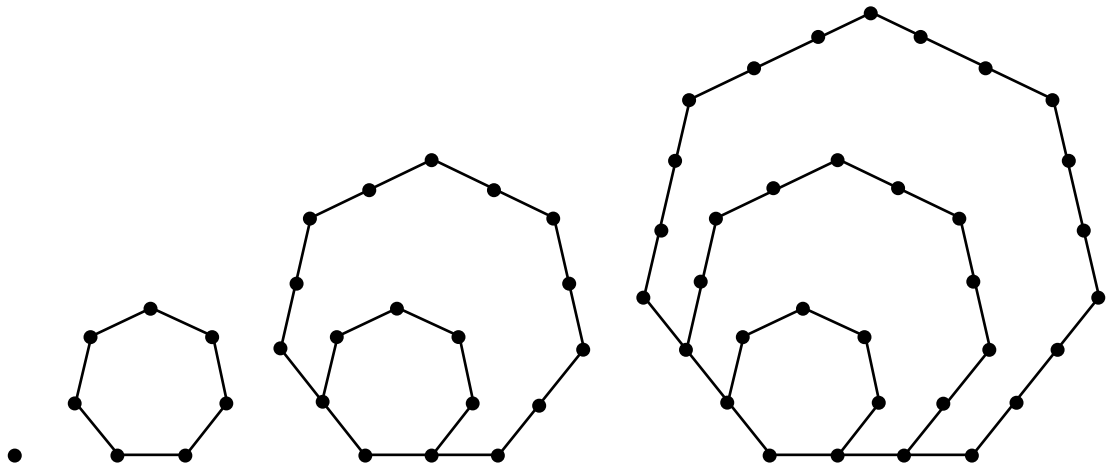


23. Bob is writing down positive integers, from left to right, starting from 1. After every 7 digits, he inserts an "@" sign. For example: 1 2 3 4 5 6 7 @ 8 9 1 0 1 1 1 @ 2 1 3 1 4 1 5 @ … and so on. What are the three digits, in exact order, immediately after the 100th "@"?

24. Consider a 6×8 board, as shown in the diagram on the right below. The tromino in the diagram on the left is made up of three 1×1 squares. In how many different ways can this tromino be placed on the board, along the grid lines, such that it covers exactly three squares?



25. As shown in the diagrams below, the first four 7-gonal numbers are 1, 7, 18 and 34, which correspond to the number of black dots in each diagram from left to right, respectively.



What is the 10th 7-gonal number?

* * *